Asta

Counterfactual Feedback, Exploration, and Self-Confirming Beliefs in Bandit Learning

Table of Contents

Introduction

Bandit feedback and the counterfactual learning problem

Information provision: randomized logging, propensities, and off-policy estimation

Forced and safe exploration in bandit learning

Counterfactual or full feedback versus observational learning

Breaking self-confirming beliefs

Introduction

In bandit learning, a learner only sees the payoff of the action it chose, so learning what would have happened under other actions is hard. Counterfactual feedback, planned exploration, and richer information can reduce this problem and help agents escape wrong but self-reinforcing beliefs. (LLM Memory)

The core issue in this area is partial feedback: when an agent chooses one action, it usually observes only that action's outcome and not the outcomes of actions it did not take. This creates a counterfactual learning problem, because the agent wants to know whether another action would have done better but does not directly see that missing information. As a result, learning can become path dependent: early choices shape later data, and later data then supports those same choices. Information provision and exploration rules try to fix this by making the missing comparisons easier to estimate, either through randomized choice, logged choice probabilities, or direct feedback about unchosen actions. This matters for observational learning too, because agents often learn from data generated by past behavior rather than by clean experiments, and such data can hide profitable actions that were rarely tried. A main theme, then, is how to design feedback and exploration so agents can learn counterfactually, make valid off-policy comparisons, and break self-confirming beliefs that persist only because the agent never sees evidence against them. (Model-Generated)

Bandit feedback and the counterfactual learning problem

Bandit learning is hard because the learner sees the result of the chosen action but not the results of the actions it did not choose. So both learning and evaluation become counterfactual: the learner must infer what would have happened under other actions from incomplete, choice-dependent data. (14 sources)

The defining feature of bandit feedback is that observation is tied to action choice: the system only observes the reward for the action it actually took, while the outcomes of the unchosen actions remain missing. This basic point is stated across contextual bandit, recommendation, and intervention settings: feedback is limited to “the particular prediction shown by the system,” and we do not get to see how the user would have responded to a different choice. (Swaminathan et al., 2015) (Joachims, 2018) (Su et al., 2018) In recommendation and other logged-decision systems, this means that learning from historical data is naturally a counterfactual inference problem, because the dataset contains realized outcomes only for executed actions. (Xiao et al., 2019) (Johansson et al., 2016)

This missing-counterfactual structure makes bandit learning different from supervised learning. In

supervised learning, labels give direct information about error on all predicted classes or at least on the realized target, but in bandit learning the learner has only a sparse and policy-shaped sample of outcomes. As a result, the learner must ask an inherently counterfactual question: how much reward would some other policy have obtained if it had been used instead of the policy that generated the log? (Sachdeva et al., 2020) (London et al., 2022) (Su et al., 2018) This is why offline policy learning from tuples like (x,a,u) requires causal and counterfactual reasoning rather than simple prediction from observed associations alone. (Hullermeier, 2021) (Bottou et al., 2012)

A second difficulty is that the data are not just incomplete but also selective. The logging policy determines which actions get observed, so some actions may have little or no data, and observed rewards can be badly biased toward the actions the system already preferred. This is the core reason why online bandits focus on exploration, while offline learning from logs focuses more on counterfactual estimation of policy value. (Ji et al., 2023) Even in newer large-action settings, the same issue appears as selection-gated logging: expected utility may depend on the whole candidate set, but the learner only observes reward for the selected item. (Tong et al., 2026)

Seen this way, bandit methods can be organized by how much feedback they assume. Some use only reward for the chosen action, others assume richer signals such as gradients, and still others assume full counterfactual rewards for all actions. The harder the feedback setting, the more the learner must reconstruct missing outcomes from design assumptions or logged information. (Durmann et al., 2026) This is also why reductions and estimators developed for partial-label or logged-bandit data are so important: they provide ways to learn despite seeing payoff for only one choice at a time. (Beygelzimer et al., 2008) (Strehl et al., 2010)

Evidence

(Swaminathan et al., 2015)

Counterfactual Risk Minimization

> *"The feedback, however, provides only partial information -"bandit feedback"-limited to the particular prediction shown by the system. The feedback for all the other predictions the system could have made is typically not known."*

(Joachims, 2018)

Deep Learning from Logged Interventions

> *"logged intervention data"*
>
> *"provides only partial-information feedback -- aka "bandit feedback" -- limited to the particular intervention chosen by the system. We don't get to see how the user would have responded, if we had chosen a different intervention."*

(Su et al., 2018)

CAB: Continuous Adaptive Blending Estimator for Policy Evaluation and Learning

> *"we only get to see the feedback for the chosen ad, but we do not observe how the user would have clicked, if the system had presented a different ad. This partial and biased nature of the feedback makes batch learning from contextual bandit feedback substantially different from typical supervised learning, where the correct label and a loss function provide full-information feedback"*
>
> *".Both learning and evaluation can be viewed as an example of counterfactual reasoning, where we would like to estimate from the historic log data how well some other policy would have performed, if we had used it instead of the policy that logged the data."*

(Xiao et al., 2019)

Model-based Constrained MDP for Budget Allocation in Sequential Incentive Marketing

> *"In real recommendation system, only feedback of executed actions is observed. Learning from logged feedback data, also called bandit problem, naturally belongs to counterfactual inference. It's necessary to consider the counterfactual risk when evaluating models and minimizing counterfactual risk becomes a reasonable objective because of incomplete observations."*

(Johansson et al., 2016)

Learning Representations for Counterfactual Inference

"In machine learning, counterfactual questions typically arise in problems where there is a learning agent which performs actions, and receives feedback or reward for that arXiv:1605.03661v2 [stat.ML] 8 Jun 2016 choice without knowing what would be the feedback for other possible choices. This is sometimes referred to as bandit feedback (Beygelzimer et al., 2010). This setup comes up in diverse areas, for example off-policy evaluation in reinforcement learning (Sutton & Barto, 1998), learning from "logged implicit exploration data" (Strehl et al., 2010) or "logged bandit feedback" (Swaminathan et al., 2015)"

(Sachdeva et al., 2020)

Off-policy Bandits with Deficient Support

"During off-policy learning, the algorithm has to address the counterfactual question of how much reward each policy in the policy space would have received, if it had been used instead of the logging policy."

(London et al., 2022)

Boosted Off-Policy Learning

"The latter learning problem is inherently counterfactual, which is why offpolicy corrections (e.g., importance weighting) are needed."

(Hullermeier, 2021)

Prescriptive Machine Learning for Automated Decision Making: Challenges and Opportunities

"More relevant from this perspective is the problem of offline (policy) learning from data of the form (x, a, u), where information about the outcome or utility u caused by a decision a made in a specific context x is given, but lacking information about counterfactuals (i.e., outcomes for decisions a = a). Variants of this problem have recently been considered in different fields. In machine learning, it is also referred to as learning from logged bandit feedback or batch learning from bandit feedback (Beygelzimer et al., 2008). This setting requires causal inference and counterfactual estimation (Bottou et al., 2012)[40]"

"As another approach for dealing with partial (observational) data, the use of propensity scores (conditional probability of a treatment/decision given the context features) has received increasing interest in the field of (medical) statistics (Rosenbaum et al., 1983)."

(Bottou et al., 2012)

Counterfactual reasoning and learning systems: the example of computational advertising

"This work shows how to leverage causal inference to understand the behavior of complex learning systems interacting with their environment and predict the consequences of changes to the system. Such predictions allow both humans and algorithms to select the changes that would have improved the system performance. This work is illustrated by experiments on the ad placement system associated with the Bing search engine."

(Ji et al., 2023)

Learning to bid and rank together in recommendation systems

"The focal point of the former is usually in how to conduct efficient exploration about the environment, while the latter is mainly in counterfactual estimation of a policy's performance."

(Tong et al., 2026)

CoFi-PGMA: Counterfactual Policy Gradients under Filtered Feedback for Multi-Agent LLMs

"This is a contextual bandit with a large action space (the action is the selected candidate) and selection-gated reward logging"

".Proposition 1 shows that under softmax routing an agent's expected utility"

"depends on the full set of candidate rewards, since the routing probabilities depend on the relative scores of all candidates. However, the observed feedback is selection-gated: we only observe r t = r(h t , a"

(Durmann et al., 2026)

Mean-based algorithms: A lower bound and regret

"One can typically distinguish algorithms based on their feedback oracle. While some algorithms only require the reward for their selected action (bandit feedback), others require more information, such as the gradients of their utility function (gradient feedback) or the counterfactual rewards for all actions (full feedback)."

(Beygelzimer et al., 2008)

The offset tree for learning with partial labels

"We present an algorithm, called the Offset Tree, for learning to make decisions in situations where the payoff of only one choice is observed, rather than all choices. The algorithm reduces this setting to binary classification, allowing one to reuse any existing, fully supervised binary classification algorithm in this partial information setting. We show that the Offset Tree is an optimal reduction to binary classification. In particular, it has regret at most (k-1) times the regret of the binary classifier it uses (where k is the number of choices), and no reduction to binary classification can do better. This reduction is also computationally optimal, both at training and test time, requiring just O(log2 k) work to train on an example or make a prediction.
Experiments with the Offset Tree show that it generally performs better than several alternative approaches."

(Strehl et al., 2010)

Learning from Logged Implicit Exploration Data

"We provide a sound and consistent foundation for the use of nonrandom exploration data in "contextual bandit" or "partially labeled" settings where only the value of a chosen action is learned. The primary challenge in a variety of settings is that the exploration policy, in which "offline" data is logged, is not explicitly known. Prior solutions here require either control of the actions during the learning process, recorded random exploration, or actions chosen obliviously in a repeated manner. The techniques reported here lift these restrictions, allowing the learning of a policy for choosing actions given features from historical data where no randomization occurred or was logged. We empirically verify our solution on two reasonably sized sets of real-world data obtained from Yahoo!."

Information provision: randomized logging, propensities, and off-policy estimation

To learn counterfactually from logged bandit data, the system usually needs some randomness in how actions were chosen and a record of those choice probabilities. Those propensities let us reweight observed outcomes so we can estimate how another policy would have performed, even though we only saw rewards for the actions actually taken. (14 sources)

A central idea in counterfactual bandit learning is that logged data become useful for off-policy evaluation only when the data contain enough randomness to support comparisons across actions. In notable work on reusing logged bandit feedback, Agarwal et al. say that the key to counterfactual reasoning from logs is randomness in the logged data, and they connect this directly to counterfactual learning and off-policy evaluation more broadly. They also note that this randomness can be added in different ways, such as randomizing actions directly or randomizing policy parameters to induce action variation. (Agarwal et al., 2017) (Bottou et al., 2012)

Once the logging policy is stochastic, the usual tool is propensity weighting: the learner records not just which action was taken, but the probability that the logging policy assigned to that action, and then reweights outcomes by the inverse of that probability. This inverse propensity scoring, or IPS, has become a standard method in counterfactual learning because it can correct for missing-not-at-random exposure and selection bias when rewards are only observed for chosen actions. The same logic has been used across recommendation, search, and learning-to-rank settings to handle selection bias and position bias in implicit feedback. (Sato et al., 2020) (Schnabel et al., 2016) (Joachims et al., 2016) (Wang et al., 2016) (Agarwal et al., 2018) (Saito et al., 2019)

This is why information provision matters so much. If a deployed system logs both actions and propensities, later learners can estimate the value of a new policy without running that policy online, at least for actions that had nonzero chance under the logger. Put simply, randomized exploration increases coverage over the action space, and unbiased counterfactual estimators then use that exploration data to estimate a candidate policy's performance. (Lecuyer et al., 2021) When exposure is randomized rather than deterministic, this also makes it possible to estimate counterfactual regret more efficiently, because one can ask not just how the observed policy did, but how regret would have changed under a different exploration rule. (Xu et al., 2022)

The flip side is that weak or missing exploration makes off-policy learning much harder. Many counterfactual bandit methods assume a stochastic logging policy precisely because otherwise some actions have no support in the data. Falke et al. stress this assumption and note that real systems often use near-deterministic argmax decisions to maximize immediate user experience, which sharply limits exploration in historical logs. They also point to work showing that learning can still be possible with deterministic logging in some cases, but only with extra smoothing or variance-control ideas, and this is a harder setting than

standard propensity-based reuse of randomized logs. (Falke et al., 2021) (Lawrence et al., 2017)

A useful way to view these methods is that information provision can happen at different levels. The system may provide randomized action choice and propensities, which support IPS-style off-policy evaluation; it may provide richer counterfactual rewards under multiple contexts, which further reduces the missing-data problem; or it may train directly with bandit or counterfactual objectives while controlling exploration. The more the system records about why an action was shown and what alternatives were possible, the easier it becomes to evaluate new policies without being trapped by the old one's selective data. (Wen et al., 2024) (Chen et al., 2025)

Evidence

(Agarwal et al., 2017)

Effective Evaluation Using Logged Bandit Feedback from Multiple Loggers

> *"e problem of re-using logged bandit feedback is o en part of counterfactual learning (Bottou et al., 2012)(Li et al., 2015)[21], and more generally can be viewed as part of o -policy evaluation in reinforcement learning (Precup et al., 2000)[20]"*

> *".e key to being able to counterfactually reason based on logged data is randomness in the logged data. Approaches di er in how randomness is being included in the policies. For example, in (Li et al., 2015) randomization is directly applied to the actions of each policy, whereas (Bottou et al., 2012) randomizes individual policy parameters to create a distribution over actions."*

(Bottou et al., 2012)

Counterfactual reasoning and learning systems: the example of computational advertising

> *"This work shows how to leverage causal inference to understand the behavior of complex learning systems interacting with their environment and predict the consequences of changes to the system. Such predictions allow both humans and algorithms to select the changes that would have improved the system performance. This work is illustrated by experiments on the ad placement system associated with the Bing search engine."*

(Sato et al., 2020)

Causality-Aware Neighborhood Methods for Recommender Systems

> *"The IPS has been gaining popularity in counterfactual learning [16](Oosterhuis et al., 2020). It has been applied to address missing not at random recommender feedback (Schnabel et al., 2016)(Saito et al., 2019), position bias in information retrieval (Wang et al., 2016)(Joachims et al., 2016)(Agarwal et al., 2018), and selection bias in bandit feedback (Bottou et al., 2012)(Zhuang et al., 2020)."*

(Schnabel et al., 2016)

Recommendations as Treatments: Debiasing Learning and Evaluation

> *"Most data for evaluating and training recommender systems is subject to selection biases, either through self-selection by the users or through the actions of the recommendation system itself. In this paper, we provide a principled approach to handle selection biases by adapting models and estimation techniques from causal inference. The approach leads to unbiased performance estimators despite biased data, and to a matrix factorization method that provides substantially improved prediction performance on real-world data. We theoretically and empirically characterize the robustness of the approach, and find that it is highly practical and scalable."*

(Joachims et al., 2016)

Unbiased Learning-to-Rank with Biased Feedback

> *"Implicit feedback (e.g., clicks, dwell times, etc.) is an abundant source of data in human-interactive systems. While implicit feedback has many advantages (e.g., it is inexpensive to collect, user centric, and timely), its inherent biases are a key obstacle to its effective use. For example, position bias in search rankings strongly influences how many clicks a result receives, so that directly using click data as a training signal in Learning-to-Rank (LTR) methods yields sub-optimal results. To overcome this bias problem, we present a counterfactual inference framework that provides the theoretical basis for unbiased LTR via Empirical Risk Minimization despite biased data. Using this framework, we derive a Propensity-Weighted Ranking SVM for discriminative learning from implicit feedback, where click models take the role of the propensity estimator. In contrast to most conventional approaches to de-biasing the data using click models, this allows training of ranking functions even in settings where queries do not repeat. Beyond the theoretical support, we show empirically that the proposed learning method is highly effective in dealing with biases, that it is robust to noise and propensity model misspecification, and that it scales efficiently. We also demonstrate the real-world applicability of our approach on an operational search engine, where it substantially improves retrieval performance."*

(Wang et al., 2016)

Learning to Rank with Selection Bias in Personal Search

> *"Click-through data has proven to be a critical resource for improving search ranking quality. Though a large amount of click data can be easily collected by search engines, various biases make it difficult to fully leverage this type of data. In the past, many click models have been proposed and successfully used to estimate the relevance for individual query-document pairs in the context of web search. These click models typically require a large quantity of clicks for each individual pair and this makes them difficult to apply in systems where click data is highly sparse due to personalized corpora and information needs, e.g., personal search. In this paper, we study the problem of how to leverage sparse click data in personal search and introduce a novel selection bias problem and address it in the learning-to-rank framework. This paper proposes a few bias estimation methods, including a novel query-dependent one that captures queries with similar results and can successfully deal with sparse data. We empirically demonstrate that learning-to-rank that accounts for query-dependent selection bias yields significant improvements in search effectiveness through online experiments with one of the world's largest personal search engines."*

(Agarwal et al., 2018)

A General Framework for Counterfactual Learning-to-Rank

> *"Implicit feedback (e.g., click, dwell time) is an attractive source of training data for Learning-to-Rank, but its naive use leads to learning results that are distorted by presentation bias. For the special case of optimizing average rank for linear ranking functions, however, the recently developed SVM-PropRank method has shown that counterfactual inference techniques can be used to provably overcome the distorting effect of presentation bias. Going beyond this special case, this paper provides a general and theoretically rigorous framework for counterfactual learning-to-rank that enables unbiased training for a broad class of additive ranking metrics (e.g., Discounted Cumulative Gain (DCG)) as well as a broad class of models (e.g., deep networks). Specifically, we derive a relaxation for propensity-weighted rank-based metrics which is subdifferentiable and thus suitable for gradient-based optimization. We demonstrate the effectiveness of this general approach by instantiating two new learning methods. One is a new type of unbiased SVM that optimizes DCG - called SVM PropDCG - and we show how the resulting optimization problem can be solved via the Convex Concave Procedure (CCP). The other is Deep PropDCG, where the ranking function can be an arbitrary deep network. In addition to the theoretical support, we empirically find that SVM PropDCG significantly outperforms existing linear rankers in terms of DCG. Moreover, the ability to train non-linear ranking functions via Deep PropDCG further improves performance."*

(Saito et al., 2019)

Unbiased Recommender Learning from Missing-Not-At-Random Implicit Feedback

> *"Recommender systems widely use implicit feedback such as click data because of its general availability. Although the presence of clicks signals the users' preference to some extent, the lack of such clicks does not necessarily indicate a negative response from the users, as it is possible that the users were not exposed to the items (positive-unlabeled problem). This leads to a difficulty in predicting the users' preferences from implicit feedback. Previous studies addressed the positive-unlabeled problem by uniformly upweighting the loss for the positive feedback data or estimating the confidence of each data having relevance information via the EM-algorithm. However, these methods failed to address the missing-not-at-random problem in which popular or frequently recommended items are more likely to be clicked than other items even if a user does not have a considerable interest in them. To overcome these limitations, we first define an ideal loss function to be optimized to realize recommendations that maximize the relevance and propose an unbiased estimator for the ideal loss. Subsequently, we analyze the variance of the proposed unbiased estimator and further propose a clipped estimator that includes the unbiased estimator as a special case. We demonstrate that the clipped estimator is expected to improve the performance of the recommender system, by considering the bias-variance trade-off. We conduct semi-synthetic and real-world experiments and demonstrate that the proposed method largely outperforms the baselines. In particular, the proposed method works better for less popular items that are less frequently observed in the training data. The findings indicate that the proposed method can better achieve the objective of recommending items with the highest relevance."*

(Lecuyer et al., 2021)

Sayer: Using Implicit Feedback to Optimize System Policies

> *"Specifically, we leverage: randomized exploration, which modifies a deployed policy to choose a random action some of the time, thereby increasing the coverage over all actions; and unbiased counterfactual estimators, which use exploration data collected from a policy to accurately estimate a candidate policy's performance."*

(Xu et al., 2022)

From Intervention to Domain Transportation: A Novel Perspective to Optimize Recommendation

> *"However, unlike the deterministic recommender, the random exposure mechanism allows efficient estimation of the counterfactual regret, i.e. what the regret would have become if an alternative exploration policy was used."*

(Falke et al., 2021)

Feedback Attribution for Counterfactual Bandit Learning in Multi-Domain Spoken Language Understanding

"A crucial assumption of most counterfactual bandit learning methods is that the logging policy is stochastic, i.e. feedback is for predictions that were sampled from π 0 and thus cover different labels for the same input. However, real-world systems, as (Lawrence et al., 2017) points out, typically use argmax predictions to deliver the best possible user experience in each case, which limits the amount of exploration that is present in historical data. In their work, they find that counterfactual learning can be possible even with fully deterministic logging policies (Lawrence et al., 2017;Lawrence and Riezler, 2018)."

(Lawrence et al., 2017)

Counterfactual Learning from Bandit Feedback under Deterministic Logging : A Case Study in Statistical Machine Translation

"The goal of counterfactual learning for statistical machine translation (SMT) is to optimize a target SMT system from logged data that consist of user feedback to translations that were predicted by another, historic SMT system. A challenge arises by the fact that risk-averse commercial SMT systems deterministically log the most probable translation. The lack of sufficient exploration of the SMT output space seemingly contradicts the theoretical requirements for counterfactual learning. We show that counterfactual learning from deterministic bandit logs is possible nevertheless by smoothing out deterministic components in learning. This can be achieved by additive and multiplicative control variates that avoid degenerate behavior in empirical risk minimization. Our simulation experiments show improvements of up to 2 BLEU points by counterfactual learning from deterministic bandit feedback."

(Wen et al., 2024)

Stochastic contextual bandits with graph feedback: from independence number to MAS number

"Such settings have also been generalized to contextual cases [BGM + 19, HZW20, DMM + 20], where counterfactual rewards could be available under different contexts."

(Chen et al., 2025)

STARS: Semantic Tokens with Augmented Representations for Recommendation at Scale

"(ii) feedback-aware training-using bandit or counterfactual objectives to align the next-item loss with business metrics while controlling for exploration."

Forced and safe exploration in bandit learning

Exploration is needed because bandit learners only learn about actions they try, but trying bad actions can hurt users or patients. A key idea is therefore to structure exploration so it is informative enough to fix blind spots while still keeping losses bounded, often by using baselines, confidence tests, or controlled forced trials. (14 sources)

A central tension in bandit learning is that the learner must sometimes choose actions that do not currently look best in order to learn whether they are actually better. This is especially clear in high-stakes domains like lending, bail, and medicine, where the algorithm only observes outcomes for the decisions it makes and therefore has to explore parts of the decision space about which it is uncertain. Without that exploration, uncertainty never gets resolved; with too much of it, the system can impose real costs on people. (Kannan et al., 2018)

This is why many papers distinguish naive exploration from safe or structured exploration. In deployed systems, fully online exploration can either fail to search enough of the action space and miss the optimum, or show users poor actions and harm their experience. Jagerman et al. propose a safe exploration scheme that starts from a baseline policy known to be acceptable, learns a candidate policy counterfactually from the baseline's behavior, and then uses high-confidence off-policy evaluation to switch only when the new policy is estimated to be at least as good as the baseline. The point is not to avoid exploration entirely, but to gate it: first prove the new policy is safe enough, then let it explore promising regions while maintaining a no-worse-than-baseline guarantee. (Jagerman et al., 2020)

A related problem is that purely counterfactual learning from old logs can fail when the logger did not explore enough. Even variance-penalized counterfactual risk minimization still struggles when importance weights explode because some actions were rarely chosen. Zenati et al. make this link explicit: under-exploration in the logging policy creates large-variance off-policy estimates, so one remedy is sequential deployment, where adaptive policies collect new data specifically to improve coverage of the action space over time. (Zenati et al., 2023) (Maurer et al., 2009) (Bottou et al., 2012)

Forced exploration is one practical answer to this coverage problem. In stationary environments, good

algorithms naturally stop trying actions that seem suboptimal, but this can be dangerous when the world changes: rarely sampled actions may improve and the learner will not notice. Gerogiannis et al. therefore argue that in nonstationary settings, extra forced exploration of rarely chosen actions is essential, though with large action spaces it must be limited to a manageable covering set rather than every possible action. (Gerogiannis et al., 2025)

The same logic appears in human and experimental work, where forced-choice trials are used to control what information the decision-maker has before free choice begins. Binz et al. describe designs in which experimenters deliberately equalize or skew prior observations across options so they can measure whether later exploration is directed toward uncertainty or produced by random choice noise. This is useful conceptually for bandits: forcing exposure can be an information design tool, not just a regret cost, because it creates the comparisons needed to separate information-seeking from mere randomness. (Binz et al., 2022) (Gershman, 2017) (Wilson et al., 2014) (Schulz et al., 2019)

There is also a stronger version of this idea in work that trains exploration policies using full or synthetic counterfactual feedback. MÊLÉE simulates contextual bandit rounds from fully labeled data and, in each round, can counterfactually evaluate all actions and compute regret estimates for each one, which are then optimized with imitation learning. This does not remove the need for exploration in real deployment, but it shows how richer training feedback can teach an agent *how* to explore better when later returned to partial-feedback environments. (Sharaf et al., 2019) (Sharaf et al., 2021)

Overall, forced exploration is best seen as a way to buy information that bandit feedback would otherwise hide. Early on, contextual bandits can behave close to randomly and incur large regret before they gather enough data, which is why exploration can feel costly in cold-start settings. But if exploration is constrained by baselines, confidence checks, or carefully chosen forced trials, it can reduce long-run regret and prevent the learner from getting stuck exploiting a policy that only looks good because alternatives were never fairly tried. (Bayley et al., 2026) (Li et al., 2010)

Evidence

(Kannan et al., 2018)

A Smoothed Analysis of the Greedy Algorithm for the Linear Contextual Bandit Problem

> *"Lending algorithms (Byrnes, 2016) observe whether individuals who were granted loans pay them back, but do not get to observe counterfactuals: would an individual not granted a loan have repaid such a loan? Algorithms which inform bail and parole decisions (Barry-Jester et al., 2015) observe whether individuals who are released go on to recidivate, but do not get to observe whether individuals who remain incarcerated would have committed crimes had they been released. Algorithms assigning drugs to patients in clinical trials do not get to observe the effects of the drugs that were not assigned to particular patients"*

> *".But they also need to explore: to make decisions which do not seem optimal according to the algorithm's current point-predictions, in order to gather more information about portions of the decision space about which the algorithm has high uncertainty."*

(Jagerman et al., 2020)

Safe Exploration for Optimizing Contextual Bandits

> *"They either do not explore the space of all possible document rankings (i.e., actions) and, thus, may miss the optimal ranking, or they present suboptimal rankings to a user and, thus, may harm the user experience. We introduce a new learning method for contextual bandit problems, Safe Exploration Algorithm (SEA), which overcomes the above drawbacks. SEA starts by using a baseline (or production) ranking system (i.e., policy), which does not harm the user experience and, thus, is safe to execute but has suboptimal performance and, thus, needs to be improved. Then SEA uses counterfactual learning to learn a new policy based on the behavior of the baseline policy. SEA also uses high-confidence off-policy evaluation to estimate the performance of the newly learned policy. Once the performance of the newly learned policy is at least as good as the performance of the baseline policy, SEA starts using the new policy to execute new actions, allowing it to actively explore favorable regions of the action space. This way, SEA never performs worse than the baseline policy and, thus, does not harm the user experience, while still exploring the action space and, thus, being able to find an optimal policy."*

(Zenati et al., 2023)

Sequential Counterfactual Risk Minimization

"Counterfactual learning from logged feedback (Bottou et al., 2012) uses only past interactions to learn a policy without interacting with the environment. Counterfactual risk minimization methods (Swaminathan & Joachims, 2015a;b) propose learning formulations using a variance penalization as in (Maurer et al., 2009) to find policies with minimal variance. Even so, counterfactual methods remain prone to large variance issues (Dudík et al., 2014). These problems may arise when the logging policy under-explores the action space, making it difficult to use importance sampling tehcniques (Owen, 2013) that are key to counterfactual reasoning. While one could collect additional data to counter this problem, our method focuses on sequential deployments (Bottou et al., 2013, see Section 6.4) to collect data obtained from adaptive policies to explore the action space."

(Maurer et al., 2009)

Empirical Bernstein Bounds and Sample-Variance Penalization

"We give improved constants for data dependent and variance sensitive confidence bounds, called empirical Bernstein bounds, and extend these inequalities to hold uniformly over classes of functions whose growth function is polynomial in the sample size n. The bounds lead us to consider sample variance penalization, a novel learning method which takes into account the empirical variance of the loss function. We give conditions under which sample variance penalization is effective. In particular, we present a bound on the excess risk incurred by the method. Using this, we argue that there are situations in which the excess risk of our method is of order 1/n, while the excess risk of empirical risk minimization is of order 1/√n. We show some experimental results, which confirm the theory. Finally, we discuss the potential application of our results to sample compression schemes."

(Bottou et al., 2012)

Counterfactual reasoning and learning systems: the example of computational advertising

"This work shows how to leverage causal inference to understand the behavior of complex learning systems interacting with their environment and predict the consequences of changes to the system. Such predictions allow both humans and algorithms to select the changes that would have improved the system performance. This work is illustrated by experiments on the ad placement system associated with the Bing search engine."

(Gerogiannis et al., 2025)

DAL: A Practical Prior-Free Black-Box Framework for Piecewise Stationary Bandits

"Forced Exploration In stationary settings, effective algorithms quickly concentrate on optimal actions for each context, rarely exploring suboptimal actions. In NS environments, however, this behavior may lead to missed changes on these rarely sampled actions, and thus, forced exploration on these actions is essential. When the action space is large or infinite, exploring all actions becomes infeasible. Therefore, DAL only does extra exploration on a finite covering set,"

(Binz et al., 2022)

Using cognitive psychology to understand GPT-3

"It is not enough to merely exploit currently available knowledge anymore, but also crucial to explore options that are unfamiliar and thereby gain information about their value. Previous research suggests that people solve this exploration-exploitation trade-off by applying a combination of two distinct strategies: directed and random exploration (Wilson et al., 2014) . Whereas directed exploration encourages the decision-maker to collect samples from previously unexplored options, random exploration strategies inject some form of stochasticity into the decision process (Gershman, 2017)(Schulz et al., 2019)"

".Forced-choice trials are used to control the amount of information available to the decision-maker. They either provide two observations for each option (equal information condition) or a single observation from one option and three from the other (unequal information condition). These two conditions make it possible to tease apart directed and random exploration by looking at the decision in the first free-choice trial. In the equal information condition, a choice is classified as random exploration if it corresponds to the option with the lower estimated mean. In the unequal information condition, a choice is classified as directed exploration if it corresponds to the option that was observed fewer times during the forced-choice trials."

(Gershman, 2017)

Deconstructing the human algorithms for exploration

"The dilemma between information gathering (exploration) and reward seeking (exploitation) is a fundamental problem for reinforcement learning agents. How humans resolve this dilemma is still an open question, because experiments have provided equivocal evidence about the underlying algorithms used by humans. We show that two families of algorithms can be distinguished in terms of how uncertainty affects exploration. Algorithms based on uncertainty bonuses predict a change in response bias as a function of uncertainty, whereas algorithms based on sampling predict a change in response slope. Two experiments provide evidence for both bias and slope changes, and computational modeling confirms that a hybrid model is the best quantitative account of the data."

(Wilson et al., 2014)

Humans Use Directed and Random Exploration to Solve the Explore–Exploit Dilemma

"All adaptive organisms face the fundamental tradeoff between pursuing a known reward (exploitation) and sampling lesser-known options in search of something better (exploration). Theory suggests at least two strategies for solving this dilemma: a directed strategy in which choices are explicitly biased toward information seeking, and a random strategy in which decision noise leads to exploration by chance. In this work we investigated the extent to which humans use these two strategies. In our "Horizon task," participants made explore– exploit decisions in two contexts that differed in the number of choices that they would make in the future (the time horizon). Participants were allowed to make either a single choice in each game (horizon 1), or 6 sequential choices (horizon 6), giving them more opportunity to explore. By modeling the behavior in these two conditions, we were able to measure exploration-related changes in decision making and quantify the contributions of the two strategies to behavior. We found that participants were more information seeking and had higher decision noise with the longer horizon, suggesting that humans use both strategies to solve the exploration– exploitation dilemma. We thus conclude that both information seeking and choice variability can be controlled and put to use in the service of exploration."

(Schulz et al., 2019)

The algorithmic architecture of exploration in the human brain.

"Balancing exploration and exploitation is one of the central problems in reinforcement learning. We review recent studies that have identified multiple algorithmic strategies underlying exploration. In particular, humans use a combination of random and uncertainty-directed exploration strategies, which rely on different brain systems, have different developmental trajectories, and are sensitive to different task manipulations. Humans are also able to exploit sophisticated structural knowledge to aid their exploration, such as information about correlations between options. New computational models, drawing inspiration from machine learning, have begun to formalize these ideas and offer new ways to understand the neural basis of reinforcement learning."

(Sharaf et al., 2019)

Meta-Learning for Contextual Bandit Exploration

"At training time (§2.3), M ÊL ÉE simulates many contextual bandit problems from fully labeled synthetic data. Using this data, in each round, M ÊL ÉE is able to counterfactually simulate what would happen under all possible action choices. We can then use this information to compute regret estimates for each action, which can be optimized using the AggreVaTe imitation learning algorithm (Ross & Bagnell, 2014)."

(Sharaf et al., 2021)

Meta-Learning Effective Exploration Strategies for Contextual Bandits

"Using this data, in each round, M ÊL ÉE is able to counterfactually simulate what would happen under all possible action choices. We can then use this information to compute regret estimates for each action, which can be optimized using the AggreVaTe imitation learning algorithm (Ross and Bagnell 2014)."

(Bayley et al., 2026)

Jump Start or False Start? A Theoretical and Empirical Evaluation of LLM-initialized Bandits

"Consequently, the agent is forced into exploration, often incurring high sample complexity and significant performance penalties before converging to a competitive strategy"

".By simultaneously balancing exploration (gathering information about user preferences) and exploitation (utilizing the information gathered to maximize some reward function), CBs optimize real-time recommendations (Li et al., 2010) and admit a sublinear finite-time regret bound under linear payoff assumptions (Chu et al., 2011). However, when CBs have yet to gather any user data, they perform essentially randomly and thus exhibit linear regret (Li et al., 2010)Auer et al., 2002)."

(Li et al., 2010)

A contextual-bandit approach to personalized news article recommendation

"Personalized web services strive to adapt their services (advertisements, news articles, etc.) to individual users by making use of both content and user information. Despite a few recent advances, this problem remains challenging for at least two reasons. First, web service is featured with dynamically changing pools of content, rendering traditional collaborative filtering methods inapplicable. Second, the scale of most web services of practical interest calls for solutions that are both fast in learning and computation.
In this work, we model personalized recommendation of news articles as a contextual bandit problem, a principled approach in which a learning algorithm sequentially selects articles to serve users based on contextual information about the users and articles, while simultaneously adapting its article-selection strategy based on user-click feedback to maximize total user clicks.
The contributions of this work are three-fold. First, we propose a new, general contextual bandit algorithm that is computationally efficient and well motivated from learning theory. Second, we argue that any bandit algorithm can be reliably evaluated offline using previously recorded random traffic. Finally, using this offline evaluation method, we successfully applied our new algorithm to a Yahoo! Front Page Today Module dataset containing over 33 million events. Results showed a 12.5% click lift compared to a standard context-free bandit algorithm, and the advantage becomes even greater when data gets more scarce."

Counterfactual or full feedback versus observational learning

When learners see outcomes for both chosen and unchosen actions, they can compare options directly and learn faster about missed opportunities. Observational or bandit learning is weaker because it only shows what happened under the chosen action, so wrong beliefs can survive simply because the learner never sees the forgone outcome. (2 sources)

A useful contrast in this literature is between **observational or bandit feedback** and **counterfactual or full feedback**. Under observational learning, the agent sees only the result of the action it took, so it must infer the value of unchosen actions indirectly from future sampling. Under full-feedback designs, the agent also sees what would have happened under the option it did not choose, which gives a direct comparison between obtained and forgone outcomes. Experimental work makes this distinction very clearly: some participants learn only from chosen outcomes, while others learn from both chosen and unchosen outcomes, that is, from explicit counterfactual feedback. (Palminteri et al., 2016) Likewise, recent work describes a "full feedback condition" in which participants receive feedback on both the selected action and the alternative action, precisely because this opens a broader space of possible learning signals than chosen-outcome feedback alone. (Rmus et al., 2025)

This difference matters because full feedback changes the learning problem itself. If the learner observes both actual and forgone rewards, it no longer has to wait until it samples the other action to learn whether it missed something better. In that sense, counterfactual feedback weakens the core bandit problem by filling in some of the missing outcomes that normally make learning path dependent. The human learning evidence is consistent with this: adults can make effective use of counterfactual information in instrumental learning when it is provided, rather than relying only on the outcome of the chosen act. (Palminteri et al., 2016)

At the same time, these studies also show that richer information does not guarantee identical use of that information across learners. Palminteri et al. report that adults efficiently integrated counterfactual information, whereas adolescents were better described by a simpler chosen-outcome learning rule. (Palminteri et al., 2016) This is important for the broader bandit theme: **providing** counterfactual information and **using** it well are different things. A system can reveal forgone outcomes, but the learner still needs a rule or model that turns those comparisons into better decisions.

So the practical contrast is simple. Observational learning leaves many beliefs self-supported because the learner only sees the consequences of what it already does. Counterfactual or full feedback gives direct evidence about roads not taken, which can speed learning, improve comparison across actions, and make it easier to detect that a currently favored action is not actually best. (Rmus et al., 2025) (Palminteri et al., 2016)

Evidence

(Palminteri et al., 2016)

The Computational Development of Reinforcement Learning during Adolescence

> *"in some contexts, participants could only learn from the outcome of their choice, whereas in other contexts they could learn from both the outcome of the chosen and the unchosen option (counterfactual learning)"*
>
> *".Behavioural analyses showed that adults learnt equally well to seek rewards and avoid punishments and also efficiently integrated counterfactual information in instrumental learning. However, adolescent behaviour displayed a different pattern. In adolescents, Bayesian model selection significantly favoured the simplest action-value algorithm (Q-learning)."*

(Rmus et al., 2025)

Generating Computational Cognitive models using Large Language Models

> *"In the other half, the participants received feedback based on their selected action, as well as counterfactual feedback from the alternative action they did not select (full feedback condition; see Figure 3A). We focused on the counterfactual condition, as it presents a broader space of possible data explanations (e.g. differences in learning based on actual and forgone rewards)."*

Breaking self-confirming beliefs

Self-confirming beliefs arise when a learner keeps choosing the same actions and therefore only sees evidence that comes from those actions. To break them, the learner needs information about actions not taken, and observational access to others' outcomes can help more than just sharing beliefs because it keeps counterfactual coverage broader over time. (1 source)

A self-confirming belief in bandit-like settings is a wrong view that survives because the learner's own choices limit what it gets to observe. If the agent avoids an action, it also avoids learning that the action may be good, so the belief sustains itself through endogenous sampling. In this sense, the problem is not just lack of data, but lack of the *right comparisons*: the learner needs evidence about options it did not pick. Park et al. study this directly and find that vicarious learning reduces self-confirming biased beliefs, measured as a stable tendency to settle on suboptimal actions, and that its main advantage comes from exposure to counterfactual information about actions not taken rather than from individual trial-and-error alone. (Park et al., 2020)

Their comparison also helps clarify what kinds of information provision are most useful for escaping these traps. Park et al. report that both individual learning and vicarious learning can improve discovery of the optimal action, but vicarious learning does better overall because it brings in information from outside the learner's own narrow choice path. They also show an important limit: simple belief sharing loses its early advantage relative to observational learning because, over time, it stops generating information on a wide range of counterfactual alternatives. Observational learning does better at resolving endogenous sampling precisely because it continues to expose the learner to others' realized experience across actions, rather than only to distilled beliefs. (Park et al., 2020)

This links back to the broader bandit theme. Counterfactual feedback, forced exploration, and observational access to others' actions all work when they widen the set of outcomes the learner can compare. A learner breaks self-confirming beliefs not merely by getting more signals, but by getting signals that speak to forgone actions. When those signals disappear, beliefs can again become self-sealing; when they are kept broad and ongoing, the learner is more likely to discover that a long-favored action is not in fact best. (Park et al., 2020)

Evidence

(Park et al., 2020)

Learning what they think vs. learning what they do: The micro-foundations of vicarious learning

> *"More importantly, we find that the two vicarious learning systems reduce SCBB, measured as a stable tendency to settle on actions other than the optimal actions, but to different degrees (see Appendix A). Even though individual learning does improve in its ability to find optimal actions with increasing (the first path to escaping self-confirming biased beliefs), across all values of , vicarious learning outperforms individual learning (indicating it is the second pathexposure to counterfactual information about actions not taken that primarily creates the advantage for either form of vicarious learning). However, belief sharing clearly loses its early advantage in terms of aiding discovery of optimal actions to observational learning. This is because it loses its ability to produce information on a wide range of counterfactual alternatives over time"*
>
> *". In other words, other's experience acquired through belief sharing is less likely to resolve the problem of endogenous sampling compared to that in observational learning."*